Inactive constraints remain strictly satisfied along a direction

Setting. Let $g_i(x) = a_i^T x - b_i$, $i = 1, \dots, p$, let $\overline{x} \in C$, and let

$$I(\overline{x}) = \{\, i : a_i^T \overline{x} - b_i = 0 \,\}$$

be the set of active constraints. Let $d \in \mathbb{R}^n$ be an arbitrary direction.

Remark. The term $t\, a_i^T d$ is *not* necessarily negative. For an inactive constraint nothing is known about the sign of $a_i^T d$ (the conditions defining $L(\overline{x}, C)$ only involve active constraints). What holds is that the *sum* $(a_i^T \overline{x} - b_i) + t\, a_i^T d$ is negative for t small enough, because the first term is a fixed negative number and the second tends to 0 as $t \to 0^+$.

Lemma 1. *There exists $\overline{t} > 0$ such that for all $t \in (0, \overline{t}]$,*

$$a_i^T(\overline{x} + td) - b_i = (a_i^T \overline{x} - b_i) + t\, a_i^T d < 0 \qquad \forall\, i \notin I(\overline{x}).$$

Proof. For each $i \notin I(\overline{x})$ define

$$s_i := -(a_i^T \overline{x} - b_i).$$

Since $\overline{x} \in C$ gives $a_i^T \overline{x} - b_i \le 0$ and $i \notin I(\overline{x})$ excludes equality, we have $s_i > 0$. Hence, for $t > 0$,

$$(a_i^T \overline{x} - b_i) + t\, a_i^T d < 0 \iff t\, a_i^T d < s_i. \tag{1}$$

Split the inactive indices into

$$J_- := \{\, i \notin I(\overline{x}) : a_i^T d \le 0 \,\}, \qquad J_+ := \{\, i \notin I(\overline{x}) : a_i^T d > 0 \,\}.$$

Case $i \in J_-$. For every $t > 0$, $t\, a_i^T d \le 0 < s_i$, so (1) holds for all $t > 0$.
Case $i \in J_+$. Define $\tau_i := s_i / (a_i^T d) > 0$. Then (1) is equivalent to $t < \tau_i$.
Now set

$$\overline{t} := \begin{cases} 1, & J_+ = \varnothing, \\ \dfrac{1}{2} \min_{i \in J_+} \tau_i, & J_+ \neq \varnothing. \end{cases}$$

Since $J_+ \subseteq \{1, \dots, p\}$ is *finite* and each $\tau_i > 0$, the minimum exists and is positive, so $\overline{t} > 0$. (With infinitely many constraints the infimum of the τ_i could be 0; this is why finiteness is used.)

Let $t \in (0, \overline{t}]$. If $i \in J_-$, (1) holds by the first case. If $i \in J_+$, then $t \le \overline{t} \le \frac{1}{2}\tau_i < \tau_i$, so (1) holds by the second case. Therefore

$$a_i^T(\overline{x} + td) - b_i < 0 \qquad \forall\, i \notin I(\overline{x}),\ \forall\, t \in (0, \overline{t}].$$

□

Remark. The factor $\frac{1}{2}$ guarantees strict inequality at the endpoint $t = \overline{t}$; at $t = \tau_i$ the constraint would be active, not strictly satisfied.